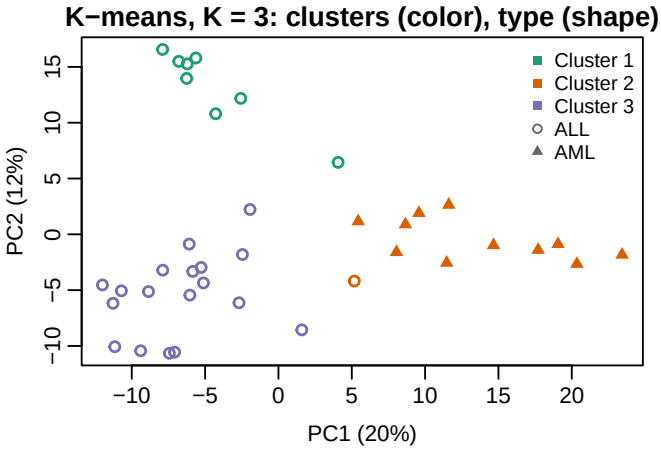
Homework 12: Clustering the Golub Leukemia Samples

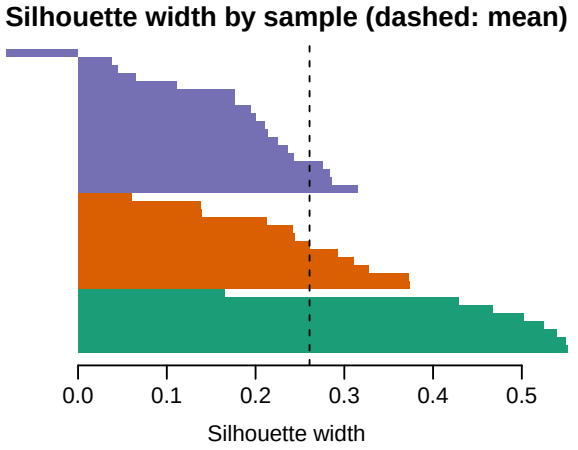
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A one-page clustering report on the Golub leukemia data (Question 1b), produced by following the skill in Appendix A; the R code is in Appendix B. **Use of AI tools.** Claude (Claude Code) drafted the skill, the R code, and the exposition, and ran the code. Every number below comes from the code output, and the numbers quoted in the text were checked against it.

Data and preprocessing. The file has 38 samples and 3051 gene-expression columns, with 0 missing values. Because $p \gg n$, I kept the 500 genes with the largest variance, standardized them, and ran PCA. The 70% rule hit the 10-PC cap; these 10 PCs explain 67.6% of the variance and form the common input for every method. The leukemia labels were not used until the final method had been chosen.

Methods and selection. I compared K-means (50 starts), Ward hierarchical clustering, and spectral clustering (symmetric 7-nearest-neighbor Gaussian graph, normalized Laplacian) for $K = 2, \ldots, 6$. For each fit I computed the mean silhouette width and the stability, the mean adjusted Rand index (ARI) over 50 refits on random 80% subsamples (seed 432). Solutions with stability ≥ 0.75 and smallest cluster ≥ 3 were admissible; each method's best admissible K by silhouette was compared, and silhouettes within 0.02 were broken by stability. The best solutions were K-means ($K = 3$: silhouette 0.261, stability 0.942); Spectral ($K = 3$: silhouette 0.259, stability 0.881); Hierarchical ($K = 3$: silhouette 0.259, stability 0.934). All 3 are within 0.02 in silhouette, so **K-means with $K = 3$** was selected for its higher stability.





Cluster	Size	ALL	AML	Mean PC1	Mean PC2	Mean silhouette	Genes with abs. SMD > 1.5
1	8	8	0	-4.4	13.3	0.466	36
2	12	1	11	12.9	-0.8	0.248	28
3	18	18	0	-6.6	-5.4	0.178	11

Interpretation. Cluster 2 contains all 11 AML samples and 1 ALL sample, and it is the only cluster with a positive mean PC1 score. Clusters 1 and 3 contain only ALL samples (8 and 18) and are separated by PC2 (mean scores 13.3 and -5.4). So the unsupervised split recovers the AML/ALL distinction except for one sample and further divides ALL into two groups. The ARI with the two known types is 0.508; merging the two ALL clusters raises it to 0.893, so the lower value comes from splitting ALL. The original study reported B-cell and T-cell ALL subtypes, so the ALL split may reflect them, but this file has no subtype labels to check it. SMD is the difference between a cluster's mean and the other samples' mean for a standardized gene; the last column counts how many of the 500 genes exceed 1.5.

Limitations. With 38 samples, the stability and silhouette values are noisy, and the three methods are nearly tied in silhouette. The results depend on the 500-gene filter, the 10-PC cap, and Euclidean distance. Cluster 3 has the weakest mean silhouette, and 1 sample has a negative silhouette width. The two-PC display shows only 32% of the variance and is not proof of distinct populations.

Appendix A: complete skill definition (`SKILL.md`)

```
---
name: cluster-select-report
description: Compare at least three clustering algorithms on a numeric data set, select one
 ↪  with label-free criteria, and write a one-page report that presents only the selected
 ↪  method. Use only when the user explicitly invokes this skill and supplies the data
 ↪  details.
disable-model-invocation: true
---

# Compare clustering methods and write a one-page report

The user supplies: the data file, the feature columns, any ID columns, any known
label columns, the language (R or Python), and the output format. Ask if any is missing.

## Rules

- Use only the feature columns for preprocessing, tuning, the number of clusters,
  and method selection. Known labels may be used only after the final method is
  chosen, to interpret its clusters.
- Fix and report every random seed. Every number in the report must come from
  code output; re-check each quoted number against the output after rendering.
- **The analysis report must fit on one printed page, including all text,
  figures, and tables.**
- **Present clustering results (assignments, cluster sizes, plots, profiles,
  label cross-tables) for only the one selected method.** The other methods may be
  named, with their selection-criterion values in one sentence or one short row
  each, to justify the choice. Do not show their partitions or plots.

## Steps

1. **Inspect.** Report n, p, missing values, and feature scales. Note whether
   p is much larger than n.
2. **Preprocess.** Handle missing values. If p is large (say p > 10n), keep the
   most variable features (for example the top 500) and say how many. Standardize
   features unless they share a meaningful common scale. Compute PCA and keep the
   smallest number of components explaining at least 70% of the variance
   (at most 10). Use these scores as the common space for every method, so the
   comparison does not depend on different inputs. PCA alone is not a clustering
   method.
3. **Candidates.** Fit at least three different algorithms, for each
   K = 2, ..., Kmax with Kmax = min(6, floor(n / 5)):
   - K-means with at least 50 random starts;
   - agglomerative hierarchical clustering with Ward linkage on Euclidean
     distance (average linkage may be added as a variant);
   - spectral clustering: Gaussian similarity on a symmetric k-nearest-neighbor
     graph (k about sqrt(n), bandwidth the median neighbor distance), normalized
     Laplacian, K-means on the row-normalized leading K eigenvectors;
   - optionally PAM or a Gaussian mixture.
4. **Criteria.** For every method and K compute, in the common PCA space:
   - mean silhouette width (Euclidean distance);
   - stability: over B >= 50 random subsamples of 80% of the observations, refit
     and compute the adjusted Rand index (ARI) with the full-data labels on the
     shared observations; report the mean;
```

```
   - the smallest cluster size.
5. **Select.** A solution is admissible if mean stability ARI >= 0.75 and the
   smallest cluster has at least max(3, 0.05 n) observations. For each method,
   take the admissible K with the largest silhouette. Choose the method with the
   largest silhouette; if two are within 0.02, prefer the higher stability, then
   the smaller K, then the simpler method (K-means, hierarchical, spectral). If no
   solution is admissible, say so and report the most stable one.
6. **Interpret the selected method.** Refit on all observations. Give cluster
   sizes and the features (or PCs) that differ most between clusters, using
   standardized mean differences. If labels were supplied, cross-tabulate
   clusters against them and give the ARI. State what the groups appear to
   represent and how confident the evidence allows one to be.
7. **Limitations.** Mention small n, the feature filter and PCA cutoff, the
   dependence on distance and scaling, stability, and that a 2-D display is not
   proof of separate populations.

## One-page report layout

- Title and one or two lines on the data and preprocessing.
- Methods and selection: name the algorithms, state the criteria and the rule, and
  give the selected method's values with one short line of comparison values.
- One figure for the selected method only, at most two panels and about 2.5 to 3
  inches tall: a scatter of the first two PCs colored by cluster (use point shape
  for known labels if supplied), plus a silhouette plot or a profile of the top
  distinguishing features.
- One small table: cluster sizes, or clusters by known label.
- Three to five sentences of interpretation, then one or two of limitations.
- Hide code on the report page; place code and any requested appendix (such as
  this skill) after a page break.
- Render, count the pages of the report section, and if it exceeds one page,
  shorten the text first, then reduce the figure. Re-render until it fits.
```

The file is 4827 bytes (limit 5,000). It was invoked explicitly with the data details: file `data/golub.csv`, features `gene_1`–`gene_3051`, ID `sample_id`, label `leukemia_type` (interpretation only), language R, output PDF.

Appendix B: R code and full comparison table

Table 2: Selection criteria for every method and K (used only to choose the method).

Method	K	Silhouette	Stability	Admissible
K-means	2	0.218	0.842	TRUE
K-means	3	0.261	0.942	TRUE
K-means	4	0.248	0.799	TRUE
K-means	5	0.242	0.706	FALSE
K-means	6	0.260	0.722	FALSE
Hierarchical	2	0.209	0.841	TRUE
Hierarchical	3	0.259	0.934	TRUE
Hierarchical	4	0.250	0.820	TRUE
Hierarchical	5	0.240	0.769	TRUE
Hierarchical	6	0.255	0.724	FALSE
Spectral	2	0.189	0.566	FALSE
Spectral	3	0.259	0.881	TRUE
Spectral	4	0.211	0.632	FALSE

Method	K	Silhouette	Stability	Admissible
Spectral	5	0.210	0.587	FALSE
Spectral	6	0.256	0.619	FALSE

```
library(cluster)   # silhouette()
library(mclust)    # adjustedRandIndex()
seed <- 432
d <- read.csv("data/golub.csv")
X <- as.matrix(d[, grep("^gene_", names(d))])
n <- nrow(X); p <- ncol(X)
n_miss <- sum(is.na(X))

# Step 2: keep the 500 most variable genes, standardize, PCA (>= 70% or at most 10 PCs)
n_keep <- 500
keep <- order(apply(X, 2, var), decreasing = TRUE)[1:n_keep]
Z <- scale(X[, keep])
pc <- prcomp(Z)
ve <- cumsum(pc$sdev^2) / sum(pc$sdev^2)
q <- min(10, which(ve >= 0.70)[1])
S <- pc$x[, 1:q]
D <- dist(S)
spectral <- function(S, K) {
  m <- nrow(S); k <- ceiling(sqrt(m))
  Dm <- as.matrix(dist(S))
  nn <- t(apply(Dm, 1, function(r) order(r)[2:(k + 1)]))
  A <- matrix(0, m, m)
  for (i in 1:m) A[i, nn[i, ]] <- 1
  A <- pmax(A, t(A))                                  # symmetric kNN graph
  sig <- median(Dm[cbind(1:m, nn[, k])])              # median k-th neighbor distance
  W <- A * exp(-Dm^2 / (2 * sig^2))
  Dh <- diag(1 / sqrt(rowSums(W)))
  L <- diag(m) - Dh %*% W %*% Dh                      # normalized Laplacian
  U <- eigen(L, symmetric = TRUE)$vectors[, m:(m - K + 1), drop = FALSE]
  U <- U / sqrt(rowSums(U^2))
  kmeans(U, K, nstart = 50)$cluster
}
fit <- function(method, S, K) switch(method,
  "K-means"      = kmeans(S, K, nstart = 50)$cluster,
  "Hierarchical" = cutree(hclust(dist(S), method = "ward.D2"), K),
  "Spectral"     = spectral(S, K))
set.seed(seed)
Kmax <- min(6, floor(n / 5)); B <- 50
crit <- NULL
for (m in c("K-means", "Hierarchical", "Spectral")) for (K in 2:Kmax) {
  cl <- fit(m, S, K)
  stab <- replicate(B, {
    i <- sort(sample(n, round(0.8 * n)))
    adjustedRandIndex(fit(m, S[i, ], K), cl[i])
  })
  crit <- rbind(crit, data.frame(method = m, K = K,
    sil = mean(silhouette(cl, D)[, 3]), stab = mean(stab), minsz = min(table(cl))))
}
min_ok <- max(3, 0.05 * n)
crit$ok <- crit$stab >= 0.75 & crit$minsz >= min_ok
best <- do.call(rbind, lapply(split(crit[crit$ok, ], crit$method[crit$ok]),
```

```
                              function(t) t[which.max(t$sil), ]))
best <- best[order(-best$sil), ]
top <- best[1, ]
near <- best[best$sil >= top$sil - 0.02, ]
near <- near[order(-near$stab, near$K), ]
sel <- near[1, ]
f3 <- function(x) sprintf("%.3f", x)
others <- best[best$method != sel$method, ]
set.seed(seed)
cl <- fit(sel$method, S, sel$K)
sil_sel <- silhouette(cl, D)
tab <- table(Cluster = cl, Type = d$leukemia_type)
ari <- adjustedRandIndex(cl, d$leukemia_type)
smd <- sapply(1:sel$K, function(k) colMeans(Z[cl == k, ]) - colMeans(Z[cl != k, ]))
n_big <- colSums(abs(smd) > 1.5)
pc_means <- apply(S[, 1:2], 2, function(z) tapply(z, cl, mean))
n_neg <- sum(sil_sel[, 3] < 0)
aml_cl <- which.max(tab[, "AML"])
all_cl <- setdiff(1:sel$K, aml_cl)
merged <- ifelse(cl == aml_cl, "AML-like", "ALL-like")
ari_merged <- adjustedRandIndex(merged, d$leukemia_type)
cols <- c("#1b9e77", "#d95f02", "#7570b3", "#e7298a", "#66a61e", "#e6ab02")[1:sel$K]
par(mfrow = c(1, 2), mar = c(3.2, 3.2, 1.6, 0.6), mgp = c(2, 0.6, 0), cex = 0.75)
plot(S[, 1], S[, 2], col = cols[cl], pch = ifelse(d$leukemia_type == "AML", 17, 1),
     lwd = 1.5, xlab = sprintf("PC1 (%.0f%%)", 100 * ve[1]),
     ylab = sprintf("PC2 (%.0f%%)", 100 * (ve[2] - ve[1])),
     main = sprintf("%s, K = %d: clusters (color), type (shape)", sel$method, sel$K))
legend("topright", c(paste("Cluster", 1:sel$K), "ALL", "AML"), bty = "n", cex = 0.85,
       col = c(cols, "grey40", "grey40"), pch = c(rep(15, sel$K), 1, 17))
o <- order(cl, -sil_sel[, 3])
barplot(sil_sel[o, 3], col = cols[cl[o]], border = NA, space = 0, horiz = TRUE,
        xlim = range(c(0, sil_sel[, 3])), xlab = "Silhouette width",
        main = "Silhouette width by sample (dashed: mean)")
abline(v = mean(sil_sel[, 3]), lty = 2)
out <- data.frame(Cluster = 1:sel$K, Size = as.vector(table(cl)),
                  ALL = tab[, "ALL"], AML = tab[, "AML"],
                  `Mean PC1` = sprintf("%.1f", pc_means[, 1]),
                  `Mean PC2` = sprintf("%.1f", pc_means[, 2]),
                  `Mean silhouette` = f3(tapply(sil_sel[, 3], cl, mean)),
                  `Genes with abs. SMD > 1.5` = n_big, check.names = FALSE)
knitr::kable(out, row.names = FALSE, align = "c")
```